Example 1

Input:

```
Order of first polynomial: 2
Enter polynomial: 1 2 3 4
Order of second polynomial: 4
Enter polynomial: 5 6 7 8 9 10
```

Output:

```
First polynomial: 4x^2 + 3x + 2 = 1
Second polynomial: 10x^4 + 9x^3 + 8x^2 + 7x + 6 = 5
Sum of polynomials: 10x^4 + 9x^3 + 12x^2 + 10x + 8 = 6
Difference of polynomials: 10x^4 + 9x^3 + 4x^2 + 4x + 4 = 4
```

Example 2

Input:

```
Order of first polynomial: 3
Enter polynomial: 2 3 4 5 6
Order of second polynomial: 3
Enter polynomial: 7 8 9 10 11
```

Output:

```
First polynomial: 6x^3 + 5x^2 + 4x + 3 = 2
Second polynomial: 11x^3 + 10x^2 + 9x + 8 = 7
Sum of polynomials: 17x^3 + 15x^2 + 13x + 11 = 9
Difference of polynomials: 5x^3 + 5x^2 + 5x + 5 = 5
```

Example 3

Input:

```
Order of first polynomial: 1
Enter polynomial: 3 4 5
Order of second polynomial: 2
Enter polynomial: 6 7 8 9
```

Output:

```
First polynomial: 5x + 4 = 3
Second polynomial: 9x^2 + 8x + 7 = 6
Sum of polynomials: 9x^2 + 13x + 11 = 9
Difference of polynomials: 9x^2 + 3x + 3 = 3
```

Example 4

Input:

```
Order of first polynomial: 2
Enter polynomial: 4 5 6 7
Order of second polynomial: 2
Enter polynomial: 8 9 10 11
```

Output:

```
First polynomial: 7x^2 + 6x + 5 = 4
Second polynomial: 11x^2 + 10x + 9 = 8
Sum of polynomials: 18x^2 + 16x + 14 = 12
Difference of polynomials: 4x^2 + 4x + 4 = 4
```

Example 5

Input:

```
Order of first polynomial: 3
Enter polynomial: 2 3 4 5 6
Order of second polynomial: 4
Enter polynomial: 7 8 9 10 11 12
```

Output:

```
First polynomial: 6x^3 + 5x^2 + 4x + 3 = 2
Second polynomial: 12x^4 + 11x^3 + 10x^2 + 9x + 8 = 7
Sum of polynomials: 12x^4 + 17x^3 + 15x^2 + 13x + 11 = 9
Difference of polynomials: 12x^4 + 5x^3 + 5x^2 + 5x + 5 = 5
```

Example 6

Input:

```
Order of first polynomial: 1
Enter polynomial: 9 8 7
Order of second polynomial: 1
Enter polynomial: 6 5 4
```

Output:

```
First polynomial: 7x + 8 = 9
Second polynomial: 4x + 5 = 6
Sum of polynomials: 11x + 13 = 15
Difference of polynomials: -3x - 3 = -3
```

Example 7

Input:

```
Order of first polynomial: 2
Enter polynomial: 3 5 7 9
Order of second polynomial: 3
Enter polynomial: 4 6 8 10 12
```

Output:

```
First polynomial: 9x^2 + 7x + 5 = 3
Second polynomial: 12x^3 + 10x^2 + 8x + 6 = 4
Sum of polynomials: 12x^3 + 19x^2 + 15x + 11 = 7
Difference of polynomials: 12x^3 + x^2 + x + 1 = 1
```

Example 8

Input:

```
Order of first polynomial: 4
Enter polynomial: 5 6 7 8 9 10
Order of second polynomial: 4
Enter polynomial: 11 12 13 14 15 16
```

Output:

```
First polynomial: 10x^4 + 9x^3 + 8x^2 + 7x + 6 = 5
Second polynomial: 16x^4 + 15x^3 + 14x^2 + 13x + 12 = 11
Sum of polynomials: 26x^4 + 24x^3 + 22x^2 + 20x + 18 = 16
Difference of polynomials: 6x^4 + 6x^3 + 6x^2 + 6x + 6 = 6
```

Example 9

Input:

```
Order of first polynomial: 3
Enter polynomial: 7 8 9 10 11
Order of second polynomial: 5
Enter polynomial: 12 13 14 15 16 17 18
```

Output:

```
First polynomial: 11x^3 + 10x^2 + 9x + 8 = 7
Second polynomial: 18x^5 + 17x^4 + 16x^3 + 15x^2 + 14x + 13 = 12
Sum of polynomials: 18x^5 + 17x^4 + 27x^3 + 25x^2 + 23x + 21 = 19
Difference of polynomials: 18x^5 + 17x^4 + 5x^3 + 5x^2 + 5x + 5 = 5
```

Example 10

Input:

```
Order of first polynomial: 2
Enter polynomial: 8 9 10 11
Order of second polynomial: 1
Enter polynomial: 12 13 14
```

Output:

```
First polynomial: 11x^2 + 10x + 9 = 8
Second polynomial: 14x + 13 = 12
Sum of polynomials: 11x^2 + 24x + 22 = 20
Difference of polynomials: -11x^2 + 4x + 4 = 4
```

Example 11

Input:

```
Order of first polynomial: 3
Enter polynomial: 3 4 5 6 7
Order of second polynomial: 3
Enter polynomial: 8 9 10 11 12
```

Output:

```
First polynomial: 7x^3 + 6x^2 + 5x + 4 = 3
Second polynomial: 12x^3 + 11x^2 + 10x + 9 = 8
Sum of polynomials: 19x^3 + 17x^2 + 15x + 13 = 11
Difference of polynomials: 5x^3 + 5x^2 + 5x + 5 = 5
```

Example 12

Input:

```
Order of first polynomial: 1
Enter polynomial: 2 3 4
Order of second polynomial: 1
Enter polynomial: 5 6 7
```

Output:

```
First polynomial: 4x + 3 = 2
Second polynomial: 7x + 6 = 5
Sum of polynomials: 11x + 9 = 7
Difference of polynomials: 3x + 3 = 3
```

Example 13

Input:

```
Order of first polynomial: 4
Enter polynomial: 6 7 8 9 10 11
Order of second polynomial: 5
Enter polynomial: 12 13 14 15 16 17 18
```

Output:

```
First polynomial: 11x^4 + 10x^3 + 9x^2 + 8x + 7 = 6
Second polynomial: 18x^5 + 17x^4 + 16x^3 + 15x^2 + 14x + 13 = 12
Sum of polynomials: 18x^5 + 28x^4 + 26x^3 + 24x^2 + 22x + 20 = 18
Difference of polynomials: 18x^5 + 6x^4 + 6x^3 + 6x^2 + 6x + 6 = 6
```

Example 14

Input:

```
Order of first polynomial: 2
Enter polynomial: 7 8 9 10
Order of second polynomial: 2
Enter polynomial: 11 12 13 14
```

Output:

```
First polynomial: 10x^2 + 9x + 8 = 7
Second polynomial: 14x^2 + 13x + 12 = 11
Sum of polynomials: 24x^2 + 22x + 20 = 18
Difference of polynomials: 4x^2 + 4x + 4 = 4
```

Example 15

Input:

```
Order of first polynomial: 3
Enter polynomial: 8 9 10 11 12
Order of second polynomial: 3
Enter polynomial: 13 14 15 16 17
```

Output:

```
First polynomial: 12x^3 + 11x^2 + 10x + 9 = 8
Second polynomial: 17x^3 + 16x^2 + 15x + 14 = 13
Sum of polynomials: 29x^3 + 27x^2 + 25x + 23 = 21
Difference of polynomials: 5x^3 + 5x^2 + 5x + 5 = 5
```

Example 16

Input:

```
Order of first polynomial: 2
Enter polynomial: 15 12 11 10
Order of second polynomial: 2
Enter polynomial: 20 18 16 14
```

Output:

```
First polynomial: 10x^2 + 11x + 12 = 15
Second polynomial: 14x^2 + 16x + 18 = 20
Sum of polynomials: 24x^2 + 27x + 30 = 35
Difference of polynomials: 4x^2 + 5x + 6 = 5
```

Example 17

Input:

```
Order of first polynomial: 1
Enter polynomial: 5 6 7
Order of second polynomial: 2
Enter polynomial: 8 9 10 11
```

Output:

```
First polynomial: 7x + 6 = 5
Second polynomial: 11x^2 + 10x + 9 = 8
Sum of polynomials: 11x^2 + 17x + 15 = 13
Difference of polynomials: 11x^2 + 3x + 3 = 3
```

Example 18

Input:

```
Order of first polynomial: 3
Enter polynomial: 12 13 14 15 16
Order of second polynomial: 3
Enter polynomial: 17 18 19 20 21
```

Output:

```
First polynomial: 16x^3 + 15x^2 + 14x + 13 = 12
Second polynomial: 21x^3 + 20x^2 + 19x + 18 = 17
Sum of polynomials: 37x^3 + 35x^2 + 33x + 31 = 29
Difference of polynomials: 5x^3 + 5x^2 + 5x + 5 = 5
```

Example 19

Input:

```
Order of first polynomial: 2
Enter polynomial: 22 23 24 25
Order of second polynomial: 4
Enter polynomial: 26 27 28 29 30 31
```

Output:

```
First polynomial: 25x^2 + 24x + 23 = 22
Second polynomial: 31x^4 + 30x^3 + 29x^2 + 28x + 27 = 26
Sum of polynomials: 31x^4 + 30x^3 + 54x^2 + 52x + 50 = 48
Difference of polynomials: 31x^4 + 30x^3 + 4x^2 + 4x + 4 = 4
```

Example 20

Input:

```
Order of first polynomial: 3
Enter polynomial: 32 33 34 35 36
Order of second polynomial: 1
Enter polynomial: 37 38 39
```

Output:

```
First polynomial: 36x^3 + 35x^2 + 34x + 33 = 32
Second polynomial: 39x + 38 = 37
Sum of polynomials: 36x^3 + 35x^2 + 73x + 71 = 69
Difference of polynomials: -36x^3 - 35x^2 + 5x + 5 = 5
```